

STOCKED

Capital Market Stock Management Application

Software Requirements Specification

Document Type	Software Requirements Specification (SRS)
Application	Stocked – Capital Market Stock Management
Version	1.0
Status	Final
Classification	Internal / Educational

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for Stocked, a Capital Market Stock Management Application. The specification serves as the authoritative reference for design, development, testing, and stakeholder communication.

1.2 Scope

Stocked is designed for educational and portfolio-management purposes. It simulates core stock market operations including stock registration, daily price tracking, transaction recording, P&L computation, real-time price feed integration, advanced portfolio analytics, multi-user support, and report exports.

1.3 Definitions & Abbreviations

Term	Definition
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement

Term	Definition
API	Application Programming Interface
REST	Representational State Transfer
P&L	Profit and Loss
FIFO	First In, First Out – cost-basis accounting method

1.4 Document Conventions

Requirements use the following prefix codes:

- FR – Functional Requirement
- NFR – Non-Functional Requirement
- DB – Database Entity
- UI – User Interface Requirement

2. Project Overview

Stocked provides a comprehensive platform for tracking and analysing stock market portfolio activity. The application targets individual investors, students, and finance enthusiasts who need a structured tool to simulate and record market operations.

2.1 Key Capabilities

- Stock counter registration and management
- Daily stock price and volume tracking
- Recording of purchase and sale transactions
- Profit / Loss calculations and portfolio valuation
- Real-time stock price integration via external APIs
- Advanced portfolio analytics and performance metrics
- Multi-user account management with role-based access
- Export of reports to PDF and Excel formats
- Interactive dashboards and visualisation charts

3. Technology Stack

Layer	Primary Technologies	Optional / Alternative
Frontend	React.js / Next.js, TypeScript	JavaScript
Visualisation	Recharts / Chart.js	D3.js

Layer	Primary Technologies	Optional / Alternative
Backend	Node.js, Express.js, REST API	—
Database	MongoDB	SQLite (lightweight deployments)
Authentication	Firebase Authentication	JWT / Session-based
External Data	Real-time stock price APIs (e.g. Alpha Vantage, Polygon.io)	Manual entry fallback
Export	PDF & XLSX generation libraries	CSV export

4. Functional Requirements

FR1 – Stock Counter Management

[FR1.1] Users shall be able to create a new stock counter providing name, ticker symbol, and optional description.

[FR1.2] Users shall be able to edit existing stock counter information.

[FR1.3] Users shall be able to view all registered stock counters with summary data.

[FR1.4] Users shall be able to delete a stock counter and its associated records.

FR2 – Daily Price Recording

[FR2.1] Users shall be able to manually record a stock's price and trading volume for any given date.

[FR2.2] The system shall display full historical price and volume records per stock.

[FR2.3] Users shall be able to update or delete incorrect price entries.

[FR2.4] When a real-time API is enabled, the system shall auto-populate daily prices from the configured feed.

FR3 – Purchase Management

[FR3.1] Users shall be able to record stock purchases specifying counter, quantity, purchase price, and date.

[FR3.2] The system shall maintain a full purchase history per stock.

[FR3.3] The system shall prevent invalid entries (e.g. negative quantities, zero price).

FR4 – Sales Management

[FR4.1] Users shall be able to record stock sales specifying counter, quantity, selling price, and date.

[FR4.2] The system shall calculate and store P&L for each sale transaction.

[FR4.3] The system shall prevent sale of quantities exceeding available holdings.

[FR4.4] Users shall be able to view full sales history including per-transaction P&L.

FR5 – Portfolio Tracking

[FR5.1] The system shall calculate and display total shares held per stock counter.

[FR5.2] The system shall calculate and display total invested capital.

[FR5.3] The system shall calculate and display realised and unrealised P&L.

[FR5.4] The portfolio summary shall refresh when new transactions or price data are recorded.

FR6 – Analytics and Reporting

[FR6.1] The system shall render price trend line charts per stock counter.

[FR6.2] The system shall render volume trend bar charts per stock counter.

[FR6.3] The system shall provide a portfolio performance graph showing cumulative P&L over time.

[FR6.4] Users shall be able to apply date-range filters to all charts and reports.

FR7 – Real-Time Stock Price Integration

[FR7.1] The system shall integrate with at least one external real-time stock price API.

[FR7.2] Users shall be able to configure their preferred price-feed provider and API credentials in settings.

[FR7.3] Price data fetched from external APIs shall be cached locally to ensure availability during outages.

[FR7.4] The system shall display the data source (live or manual) and last-updated timestamp for all price data.

[FR7.5] If the external API is unavailable, the system shall fall back to cached data and notify the user.

FR8 – Advanced Portfolio Analytics

[FR8.1] The system shall compute portfolio-level metrics including total return, annualised return, and volatility.

[FR8.2] The system shall provide stock-level allocation breakdowns by portfolio value and holding percentage.

[FR8.3] The analytics module shall support performance benchmarking across multiple stock counters.

[FR8.4] Users shall be able to set performance targets and view progress tracking against those targets.

FR9 – Multi-User Support

[FR9.1] The system shall support multiple independent user accounts each with isolated portfolio data.

[FR9.2] Users shall register and authenticate via a secure login mechanism.

[FR9.3] Role-based access control shall prevent any user from viewing or modifying another user's data.

[FR9.4] An administrator role shall have the ability to manage user accounts.

FR10 – Report Export (PDF and Excel)

[FR10.1] Users shall be able to export any portfolio summary, transaction history, or analytics view as a PDF document.

[FR10.2] Users shall be able to export tabular data (transactions, holdings, price history) as an Excel (.xlsx) file.

[FR10.3] Exported filenames shall include the stock counter, report type, and date range.

[FR10.4] PDF exports shall include Stocked branding, generation timestamp, and user identifier.

5. Non-Functional Requirements

ID	Category	Requirement
NFR1.1	Performance	API responses shall complete within 2 seconds under normal load.
NFR1.2	Performance	Real-time price data shall refresh at intervals no greater than 60 seconds when an active feed is configured.
NFR2.1	Usability	The interface shall be fully responsive across desktop, tablet, and mobile viewports.
NFR2.2	Usability	Navigation shall be intuitive; a new user shall be able to record a transaction within 3 minutes of first use.
NFR3.1	Reliability	All transaction inputs shall be validated server-side before persistence.
NFR3.2	Reliability	Cached price data shall ensure report generation remains available during API outages.
NFR4.1	Security	All API endpoints shall validate and sanitise user input to prevent injection attacks.
NFR4.2	Security	User authentication shall be enforced across all protected routes when multi-user mode is active.
NFR4.3	Security	API keys for external price feeds shall be stored securely in environment variables and never exposed client-side.
NFR5.1	Scalability	The architecture shall support horizontal scaling to accommodate growing multi-user workloads.

6. Database Design

The following entities form the core data model for Stocked. Relationships are enforced via foreign-key references at the application layer.

Entity	Key Fields	Notes
Users	id, email, passwordHash, role	Multi-user support. Roles: admin user.
Stocks	id, userId, name, symbol	Scoped per user. Symbol unique per user.
DailyPrices	id, stockId, date, price, volume, source	source: manual api. Composite unique key (stockId, date).
Purchases	id, stockId, quantity, purchasePrice, purchaseDate	Supports average cost or FIFO basis.
Sales	id, stockId, quantity, sellPrice, saleDate, profitLoss	profitLoss calculated and stored at time of sale.
ExportLogs	id, userId, exportType, generatedAt, filename	Audit trail for PDF/Excel exports.

7. User Interface Requirements

Screen	Requirements
Dashboard	Portfolio summary cards (total value, total P&L, top performers). Recent transactions list. Real-time price ticker when live feed is active.
Stocks Page	Add / Edit / Delete stock counters. Search and filter. Real-time price badge per counter when API is enabled.
Prices Page	Manual price and volume entry form. Data-source indicator (manual vs. live). Historical price table with pagination.
Transactions	Purchase and sale entry forms with validation feedback. Transaction history with date-range filtering. Per-transaction P&L display.
Analytics	Line charts (price trends), bar charts (volume), portfolio performance graphs. Advanced metrics panel. Export to PDF / Excel buttons.
Settings	API key configuration for price feeds. Refresh-interval preferences. User account management (multi-user mode).

8. Assumptions and Constraints

8.1 Assumptions

- Manual data entry remains available as a fallback when external API feeds are unavailable or not configured.

- P&L calculations use average purchase cost by default; FIFO may be implemented as an optional configuration.
- The application targets modern evergreen browsers (Chrome, Firefox, Edge, Safari).
- Users are responsible for obtaining valid API credentials for any third-party price services they configure.

8.2 Constraints

- Real-time price data accuracy depends on the reliability and terms of the configured third-party API provider.
- Export file sizes for large datasets may be subject to browser or server memory limits.
- Stocked is classified as an educational application and is not intended for regulated financial advisory use.

9. Revision History

Version	Date	Author	Summary
1.0	Initial	Project Team	Initial SRS draft – Capital Market Stock App
2.0	Current	Project Team	Renamed to Stocked. FR7-FR10 added (real-time API, advanced analytics, multi-user, PDF/Excel export). All sections refined. Users, ExportLogs entities added.